

1.11.2 Testing Methods (Usability, A/B, Automated)

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CS XII

Chapter 1: Computer Systems (HCI)

Lecture 15 of 15

★ Golden

1.11.2 Testing Methods (Usability, A/B, Automated) ★ Golden

LEARN IT

Testing involves observing how the system behaves under specific conditions. Learn all seven methods. The three named in the heading — usability, A/B and automated — are the ones examiners ask most, but a complete answer also names the others.

1. **Usability testing:** observing real users while they perform realistic tasks, to measure efficiency and satisfaction. If the user gets stuck, designers know where to fix the UI.
2. **Task-based testing:** assigning a specific task (for example, "Find a product and add it to cart") and strictly measuring **completion time**, **success rate** and **number of errors**.
3. **Automated audits:** software bots scan websites to detect technical issues such as broken links, slow loading speeds or low colour contrast.
4. **A/B testing:** comparing two versions of an interface (A and B). Half the users see A, half see B. The one with more clicks or sign-ups wins.
5. **Compatibility testing:** verifying the system on different browsers (Chrome, Safari), devices (mobiles, tablets, PCs) and operating systems.
6. **Performance and load testing:** simulating high traffic (for example 10,000 users logging in at once) to see if speed drops or the server crashes.
7. **Keyboard and screen-reader walkthroughs:** navigating using *only* the keyboard (Tab/Enter) or assistive technologies, to check accessibility for users with disabilities.

Task-based

"Add to cart"

Time: 12 s

Success: Yes

A/B test

Version A | Version B

half / half users

more clicks wins

Automated audit

Bot scanning...

3 broken links

Low contrast

Three golden testing pictures: timed tasks, A versus B, and a bot that finds broken links and weak contrast.

WORKED EXAMPLE

Worked example: test JazzCash "Send Money" and a school portal with numbers.

Usability testing. Sit beside five students. Task: "Send Rs 100 to this test number." Do not help. Note where they pause. If three people look for Confirm at the bottom and it is at the top, that is your UI fix.

Task-based testing (numbers). Same task for 10 students.

- Success: 8 of 10 finish = success rate $(8 \div 10) \times 100 = 80\%$.
- Times: 12, 14, 11, 40, 13, 12, 15, 50, 14, 12 seconds. The two slow times (40, 50) are the students who got stuck — they also made errors.
- Errors: extra taps, wrong field. Count them. High errors + high time = poor HCI, even if most people eventually succeed.

A/B testing. Version A: green "Send" at the top. Version B: large orange "Send" under the amount. 200 users each. If B gets 140 completed payments and A gets 90, B wins. You keep B.

Automated audit. A bot reports 3 broken "Fee challan" links and low contrast on the portal's grey-on-grey marks table. Fix those before the board-exam rush.

Load testing. On result day, 10,000 students log in at 9:00 a.m. If the server crashes, performance testing failed — the UI never even appears.

Keyboard walkthrough. Unplug the mouse. Can a student Tab to "Download date sheet" and press Enter? If focus jumps randomly, accessibility testing failed.

REMEMBER IT

Task-based three numbers: Time, Success, Errors. **A/B:** two versions, split users, the higher clicks/sign-ups win. **Bot:** links, speed, contrast. **Load:** many users at once. **Keyboard:** Tab and Enter only.

COMMON MISTAKE

Mistake: Calling A/B testing "showing both designs to the same user and asking which is prettier".

Fix: Half the users see only A, half see only B. You measure behaviour (clicks, sign-ups), not a taste vote. Also: an automated audit does not replace watching real users — bots cannot see that a student is confused.

اردو میں سمجھیں

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EXAM TIP

Golden 8-mark question: "Explain testing methods used in HCI." Write seven short paragraphs or a seven-row table. For full marks, define usability testing (observe real users on realistic tasks), give the three task-based measures, describe automated audits with the three bot findings (broken links, slow load, low contrast), explain A/B with the half-and-half rule, then add compatibility, load (10,000 users), and keyboard/screen-reader walkthroughs. If the paper asks only "usability, A/B and automated", go deep on those three and still mention the rest in one closing line.

CHECK YOURSELF

1. Name the three measurements in task-based testing.
2. In A/B testing, who sees version A?
3. Give two issues an automated audit can detect.

Answers: 1. Completion time, success rate, number of errors. 2. Half of the users (the other half see B). 3. Any two of: broken links, slow loading, low colour contrast.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Usability testing is:

- a. Watching people do real tasks
- b. Only pinging servers
- c. Only unit tests of Python
- d. Only load tests of CPUs

MCQ 2. A/B testing compares:

- a. Two versions on similar users
- b. NAND vs NOR only
- c. Mean vs mode only
- d. TCP vs UDP only

MCQ 3. Automated UI tests are good for:

- a. Regressions of known flows
- b. Discovering new user feelings
- c. Replacing all interviews
- d. Measuring joy perfectly

MCQ 4. Think-aloud means:

- a. Users speak while they try the task
- b. The computer reads NAND
- c. SQL dumps
- d. Sprinters shout

MCQ 5. A/B needs:

- a. A clear metric (clicks, success)
- b. Random colours with no goal
- c. $n=1$
- d. No ethics

MCQ 6. This is ★: name the method, when to use it, one limit.

- a. Yes — that is the 5-mark recipe
- b. No — only definitions
- c. No — write code
- d. No — draw OSI

Extra short questions

1. Define usability testing, A/B, automated testing.
2. One limit of each.
3. When A/B is overkill in XI/XII class.
4. What do you record in a usability test?

Extra long question

1. Design a usability test + a tiny A/B for a “download slip” button (blue text vs big labelled button). Tasks, n, metric, ethics. ★ Golden.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Usability: observe tasks. A/B: two live variants. Automated: scripts click through.
2. Usability: small n, lab effect. A/B: needs traffic, can harm half the users. Auto: misses confusion that does not throw errors.
3. Five classmates on a Figma prototype is enough — no live traffic.
4. Task, success, time, errors, quotes, where they hesitated.

Long

1. Task: get the slip. $n=10$ (5+5). Metric: success and time. Ethics: consent, no marks attached. Predict big button wins. Also one automated check that the link still works.